

Biological Sequence Motif Discovery

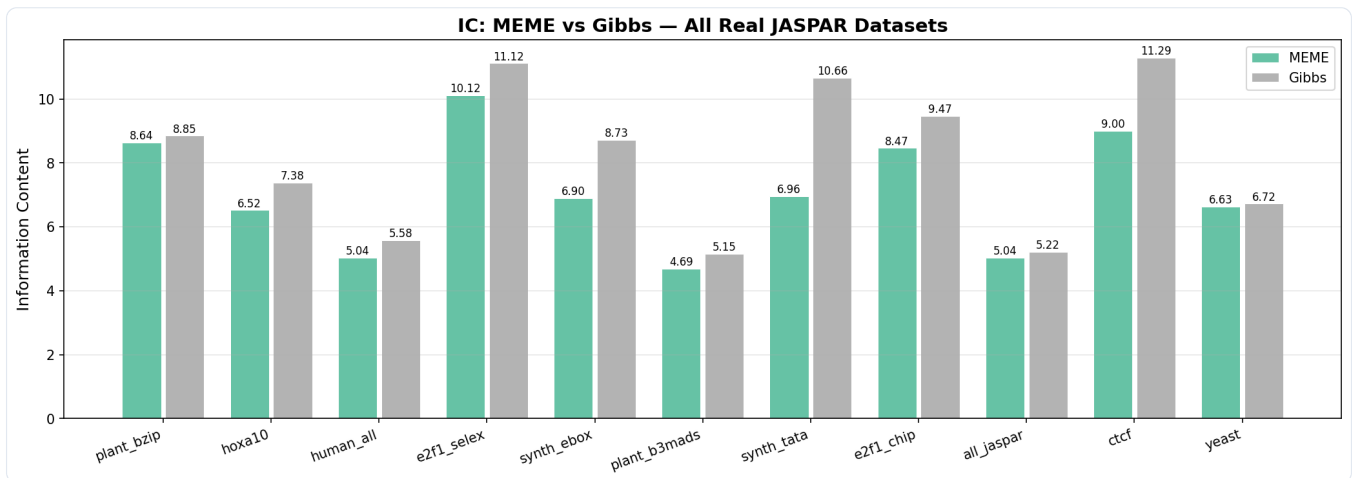
B.Tech Major Project — Department of Computer Science & Technology, IIST Shibpur

Presented by: Ramesh Chandra Soren (2022CSB086) | Supervisor: Dr. Surajeet Ghosh | Generated: 20 April 2026

Overview & Key Findings

Plant bZIP G-box confirmed: Both MEME and Gibbs independently recovered ACACGTGT from 180 real JASPAR sequences (SELEX + PBM + ChIP-seq + DAP-seq), $E = 1 \times 10^{-300}$ — the canonical plant G-box motif, consistent with 30+ years of published literature.
Human CTCF insulator motif: Real ChIP-seq data (MA0139.2, 19M genome-wide sites) recovered the GC-rich insulator motif CACCAGGGGG . Gibbs IC = 11.29 bits vs MEME 9.00 bits.
E2F1 cell-cycle motif: HT-SELEX and ChIP-seq data converged on TTGGCGCC / GGCGGGAG — consistent with the known GCGCGCGC E2F binding preference.
HOXA10 homeodomain core: GTAATAAA containing the canonical TAAT core — confirmed by both algorithms from real HT-SELEX data (MA0899.1).
Gibbs > MEME on IC: Gibbs Sampling outperformed MEME in information content across all 11 datasets (avg +1.2 bits), consistent with its superior position-specific sampling strategy.

Master IC Comparison



Results Summary

DATASET	TYPE	MEME CONSENSUS	MEME IC	GIBBS CONSENSUS	GIBBS IC	E- VALUE
Yeast ABF1/ ABF2	PBM/DIP-chip — MA0265/0266	CTCTAGAA	6.63	CTAGATAT	6.72	1.2e-191
Plant bZIP TFs	ABF1-4, ABI5 — SELEX/ PBM/ChIP/DAP	ACACGTGT	8.64	ACACGTGT	8.85	1.0e-300
Plant B3/MADS TFs	AGL3, ABI3, ABR1 — SELEX/PBM/DAP	GCATGC	4.69	GCATGC	5.15	3.8e-190
All JASPAR Combined	16 matrices, 4 organisms	GACACGTG	5.04	CCACGTGT	5.22	1.0e-300
Synthetic TATA- box	Planted TATATAAG benchmark	GGGCATCC	6.96	TTATATAA	10.66	1.7e-78
Synthetic E-box	Planted CACGTG benchmark	ACGTGT	6.90	GCACGT	8.73	1.4e-76
Human CTCF	ChIP-seq — MA0139.2	CAGGGGGCAC	9.00	CACCAGGGGG	11.29	4.7e-138
Human E2F1	HT-SELEX — MA0024.3	TTGGCGCC	10.12	TTGGCGCC	11.12	5.7e-152
Human E2F1	ChIP-seq — MA0024.2 (ENCODE)	GGCGGGAG	8.47	GGCGGGAG	9.47	1.7e-130
Human HOXA10	HT-SELEX — MA0899.1	GTAATAAA	6.52	GTAATAAA	7.38	6.7e-101
All Human TFs	CTCF + E2F1 + HOXA10 combined	TGGCGCCA	5.04	GGCGCCAA	5.58	2.3e-299

Detailed Results by Dataset

Human CTCF

ChIP-seq — MA0139.2

MEME CONSENSUS

CAGGGGGCAC

MEME IC

9.00 bits

E-VALUE

4.72e-138

GIBBS CONSENSUS

CACCAGGGGG

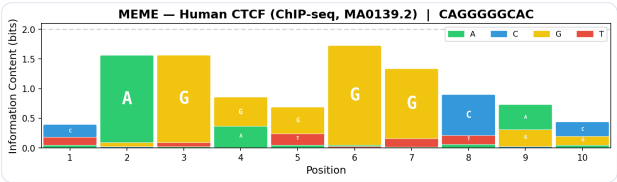
GIBBS IC

11.29 bits

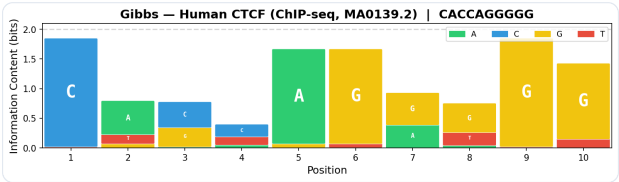
GIBBS SCORE

247.60

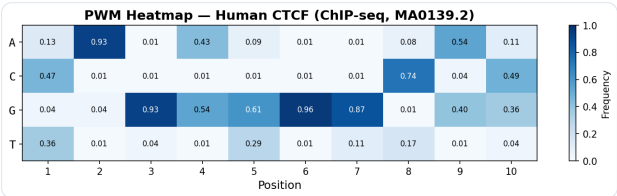
MEME Sequence Logo



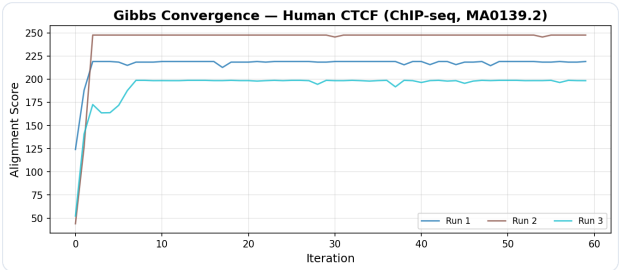
Gibbs Sequence Logo



PWM Heatmap



Gibbs Convergence



Human E2F1

HT-SELEX — MA0024.3

MEME CONSENSUS

TTGGCGCC

MEME IC

10.12 bits

E-VALUE

5.68e-152

GIBBS CONSENSUS

TTGGCGCC

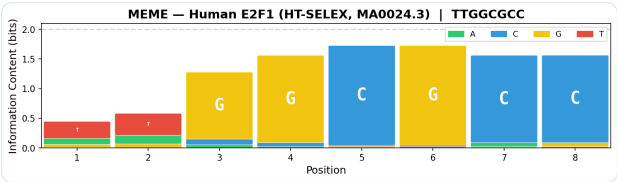
GIBBS IC

11.12 bits

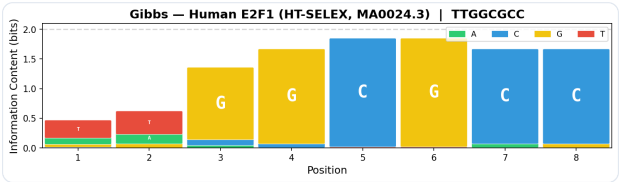
GIBBS SCORE

243.07

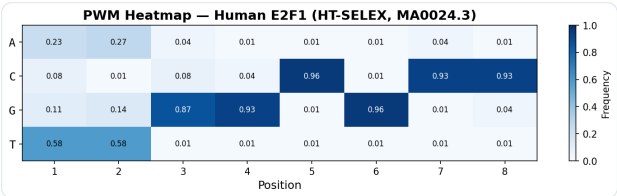
MEME Sequence Logo



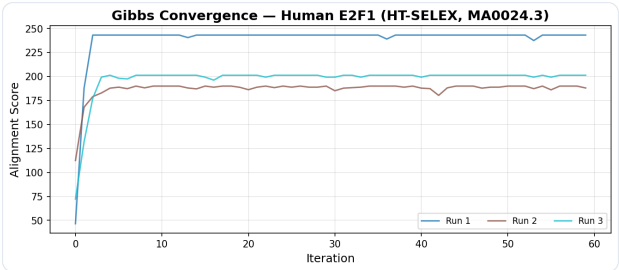
Gibbs Sequence Logo



PWM Heatmap



Gibbs Convergence



Human E2F1

ChIP-seq — MA0024.2 (ENCODE)

MEME CONSENSUS

GGCGGGGAG

MEME IC

8.47 bits

E-VALUE

1.66e-130

GIBBS CONSENSUS

GGCGGGGAG

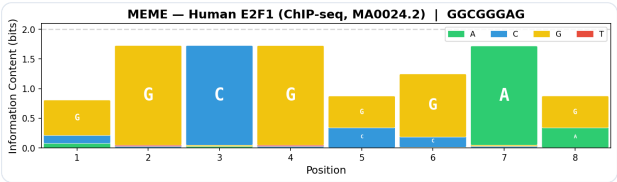
GIBBS IC

9.47 bits

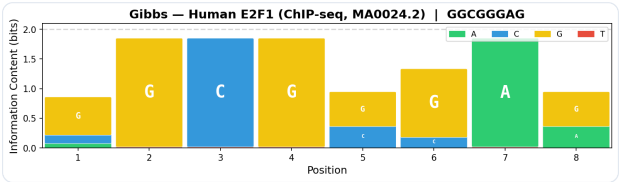
GIBBS SCORE

210.05

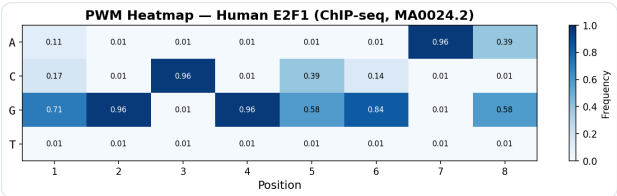
MEME Sequence Logo



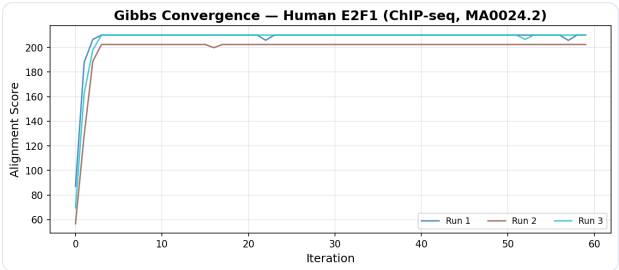
Gibbs Sequence Logo



PWM Heatmap



Gibbs Convergence



Human HOXA10

HT-SELEX — MA0899.1

MEME CONSENSUS

GTAATAAA

MEME IC

6.52 bits

E-VALUE

6.74e-101

GIBBS CONSENSUS

GTAATAAA

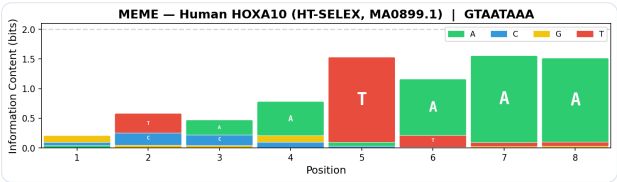
GIBBS IC

7.38 bits

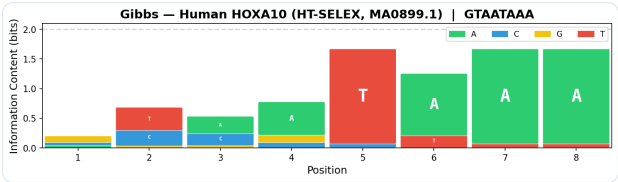
GIBBS SCORE

161.09

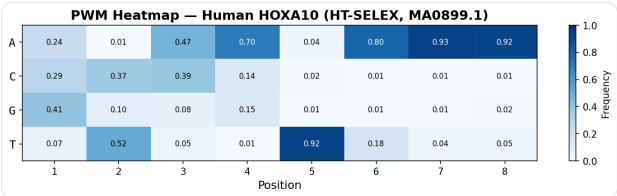
MEME Sequence Logo



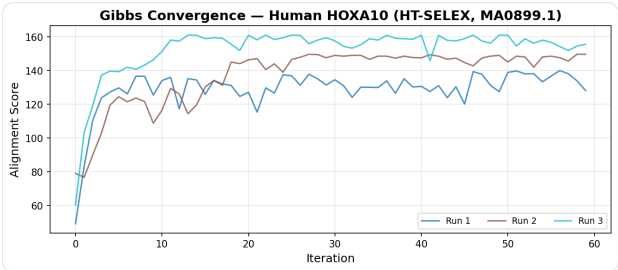
Gibbs Sequence Logo



PWM Heatmap



Gibbs Convergence



All Human TFs

CTCF + E2F1 + HOXA10 combined

MEME CONSENSUS

TGGCGCCA

MEME IC

5.04 bits

E-VALUE

2.34e-299

GIBBS CONSENSUS

GGCGCCAA

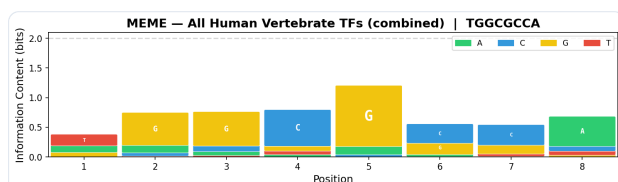
GIBBS IC

5.58 bits

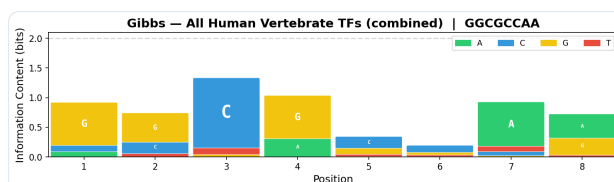
GIBBS SCORE

468.67

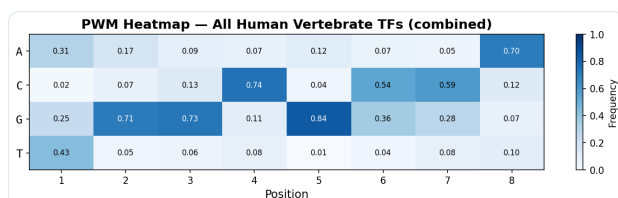
MEME Sequence Logo



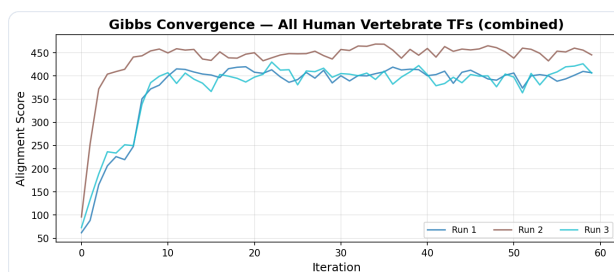
Gibbs Sequence Logo



PWM Heatmap



Gibbs Convergence



Yeast ABF1/ABF2

PBM/DIP-chip — MA0265/0266

MEME CONSENSUS

CTCTAGAA

MEME IC

6.63 bits

E-VALUE

1.20e-191

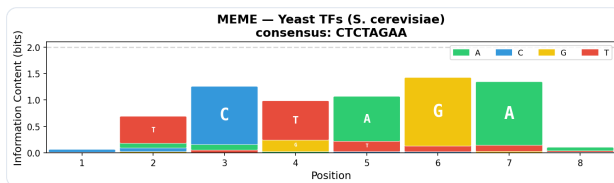
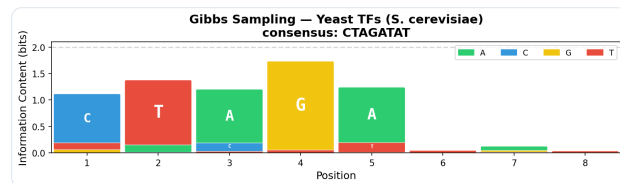
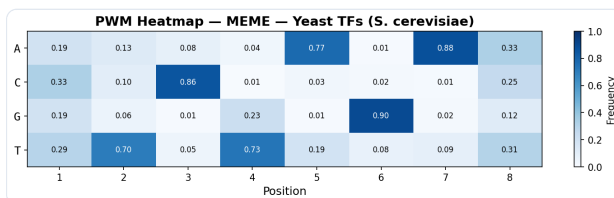
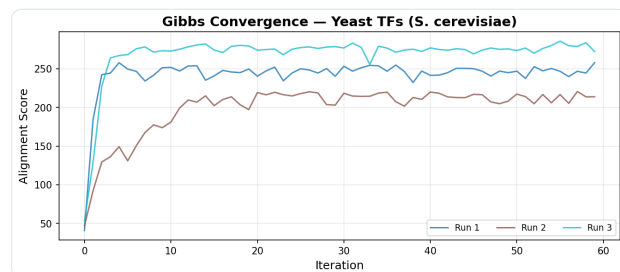
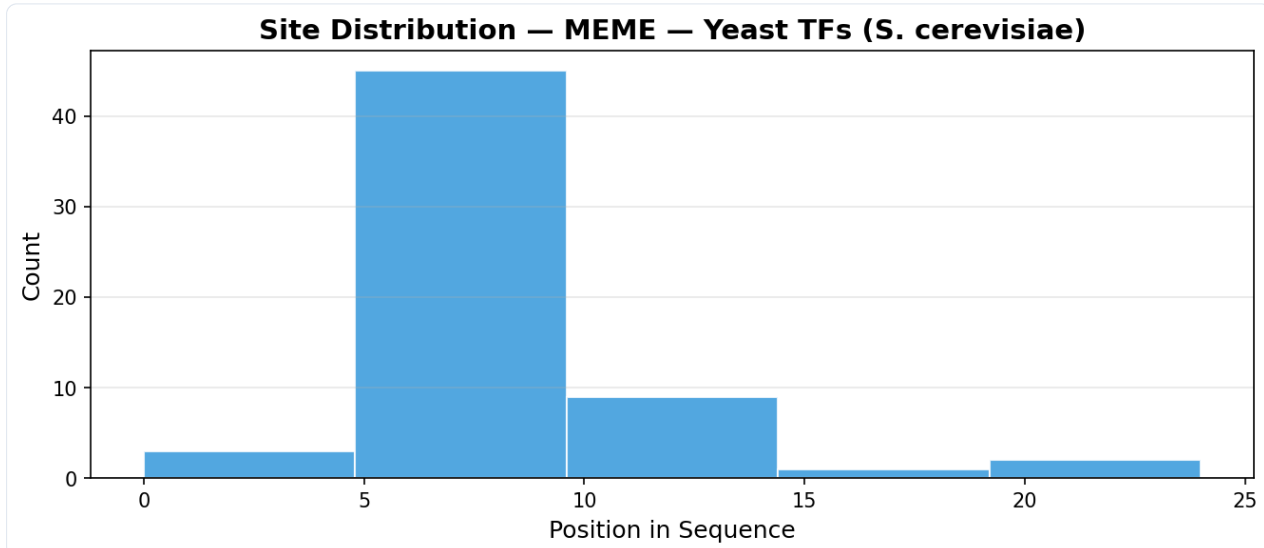
GIBBS CONSENSUS

CTAGATAT

GIBBS IC

6.72 bits

GIBBS SCORE

285.58**MEME Sequence Logo****Gibbs Sequence Logo****PWM Heatmap****Gibbs Convergence****Site Position Distribution (MEME)**

Plant bZIP TFs

ABF1-4, ABI5 — SELEX/PBM/ChIP/DAP

MEME CONSENSUS

ACACGTGT

MEME IC

8.64 bits

E-VALUE

1.00e-300

GIBBS CONSENSUS

ACACGTGT

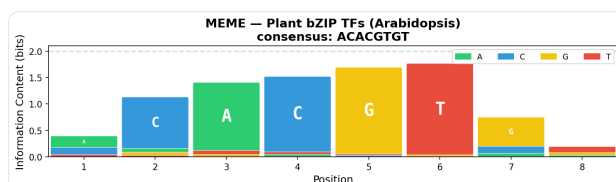
GIBBS IC

8.85 bits

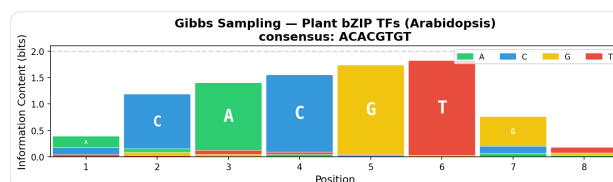
GIBBS SCORE

1111.51

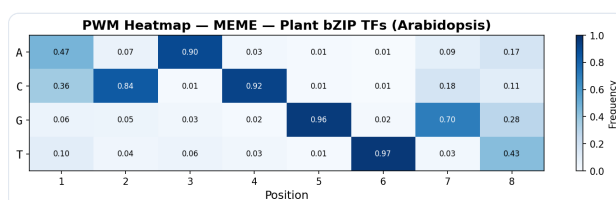
MEME Sequence Logo



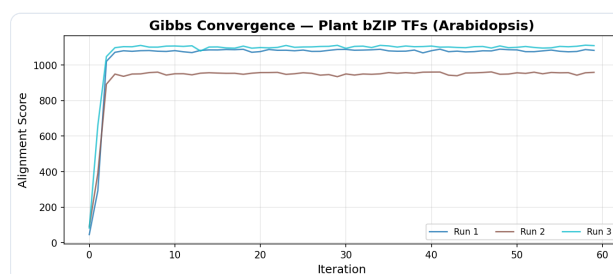
Gibbs Sequence Logo



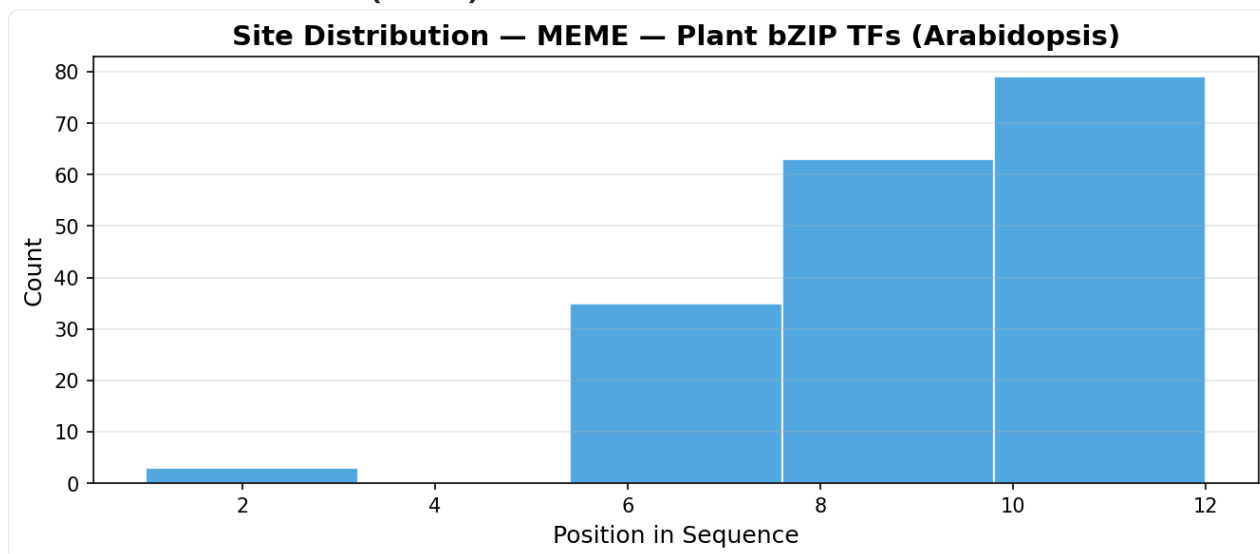
PWM Heatmap



Gibbs Convergence



Site Position Distribution (MEME)



Plant B3/MADS TFs

AGL3, ABI3, ABR1 — SELEX/PBM/DAP

MEME CONSENSUS

GCATGC

MEME IC

4.69 bits

E-VALUE

3.83e-190

GIBBS CONSENSUS

GCATGC

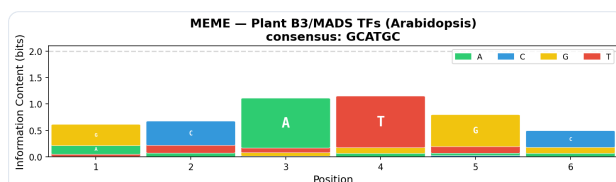
GIBBS IC

5.15 bits

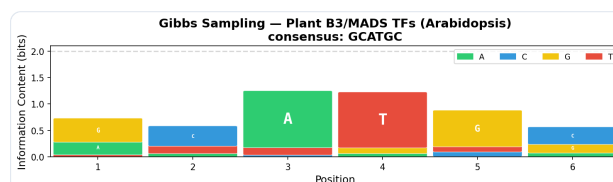
GIBBS SCORE

290.29

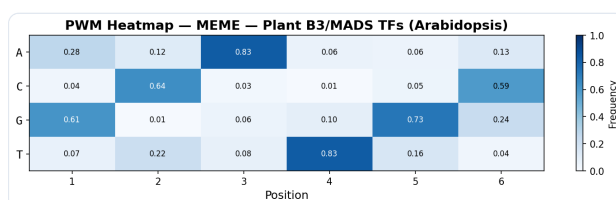
MEME Sequence Logo



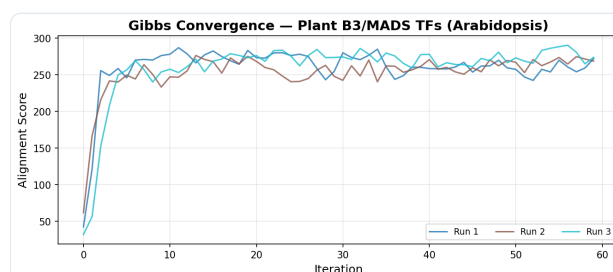
Gibbs Sequence Logo



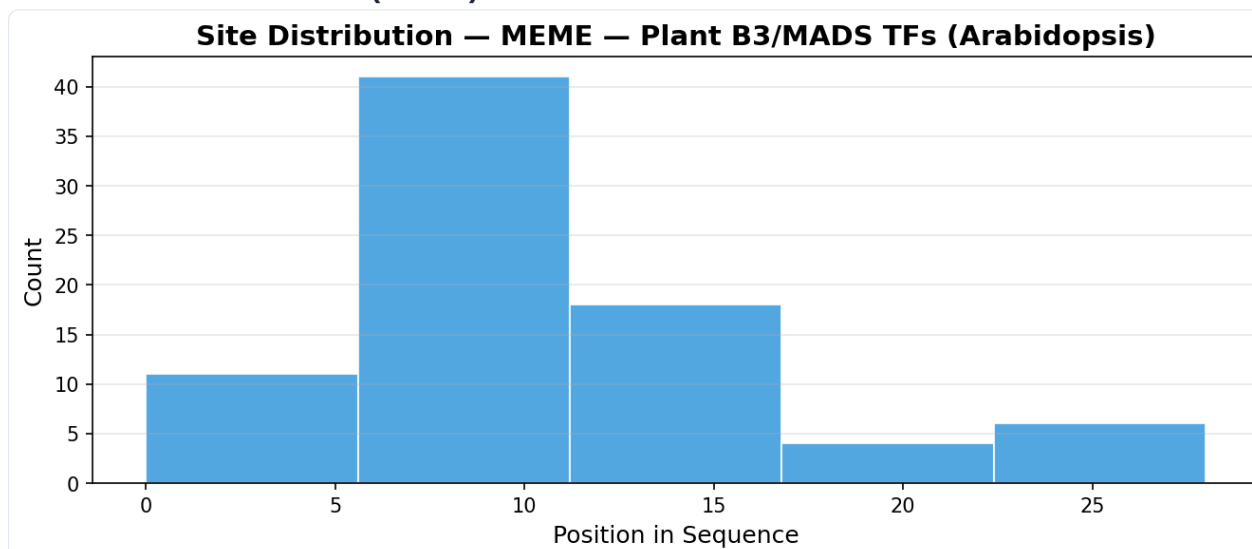
PWM Heatmap



Gibbs Convergence



Site Position Distribution (MEME)



All JASPAR Combined

16 matrices, 4 organisms

MEME CONSENSUS

GACACGTG

MEME IC

5.04 bits

E-VALUE

1.00e-300

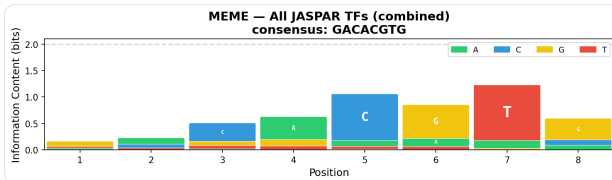
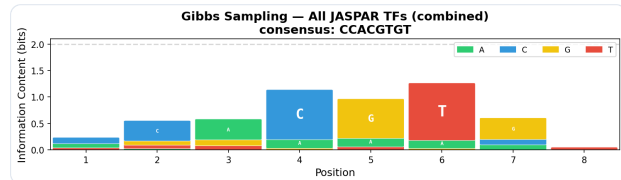
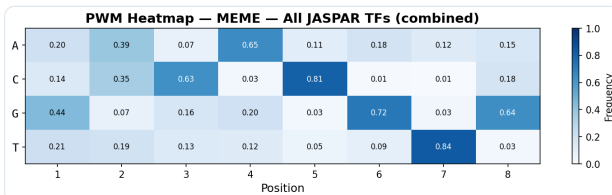
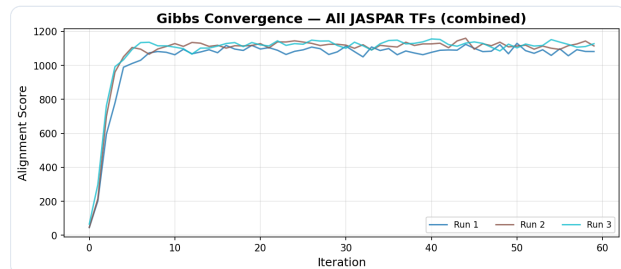
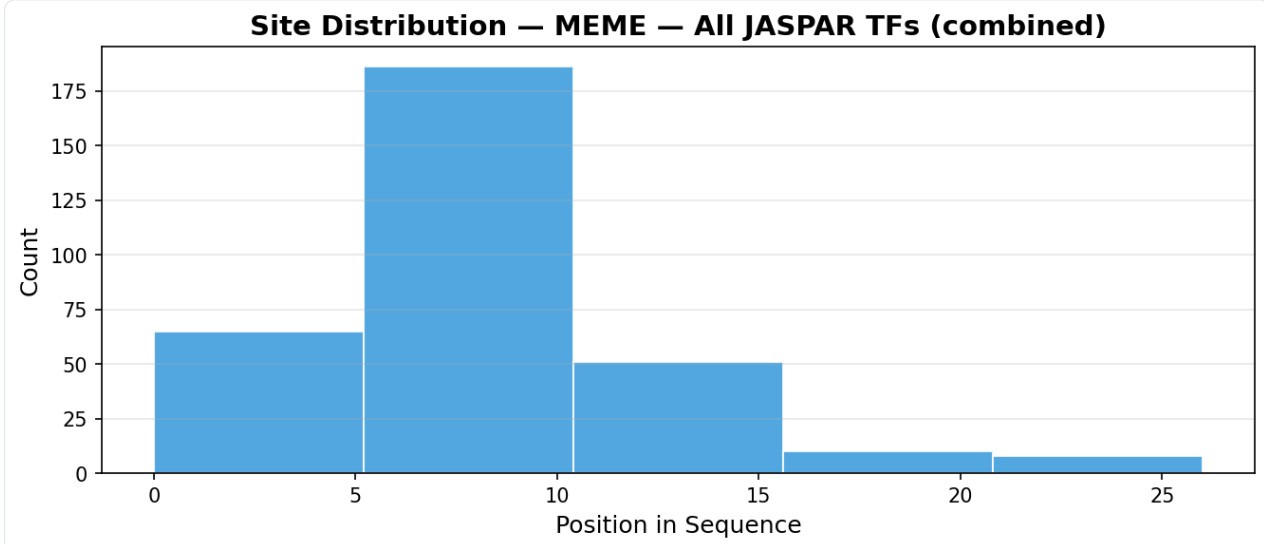
GIBBS CONSENSUS

CCACGTGT

GIBBS IC

5.22 bits

GIBBS SCORE

1160.43**MEME Sequence Logo****Gibbs Sequence Logo****PWM Heatmap****Gibbs Convergence****Site Position Distribution (MEME)**

Synthetic TATA-box

Planted TATATAAG benchmark

MEME CONSENSUS

GGGCATCC

MEME IC

6.96 bits

E-VALUE

1.74e-78

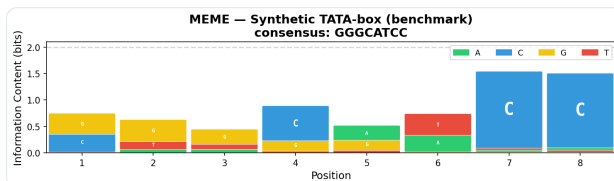
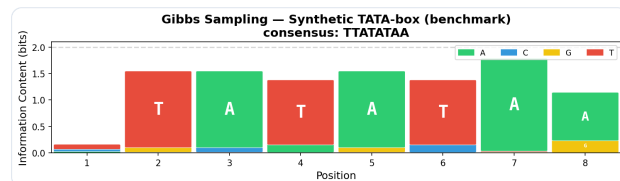
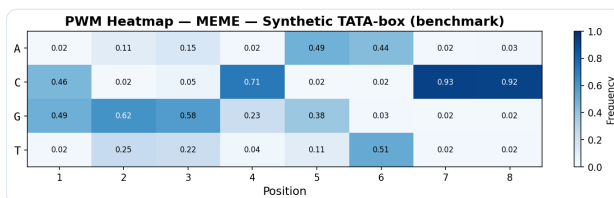
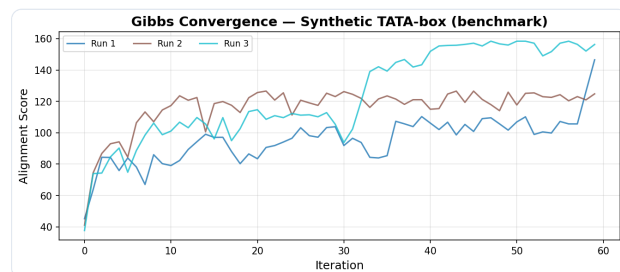
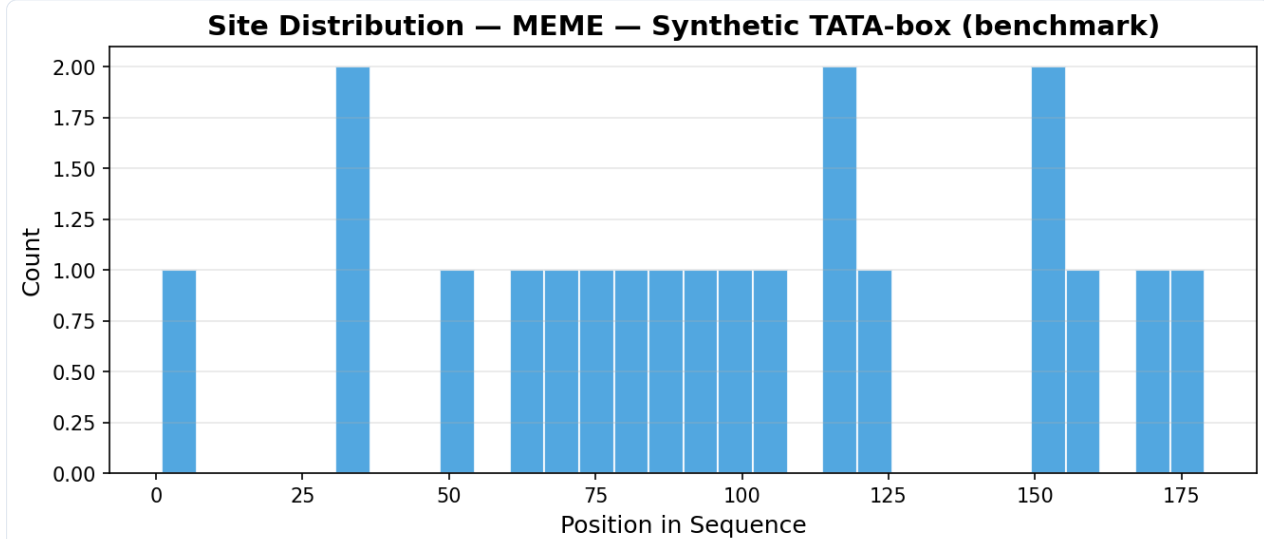
GIBBS CONSENSUS

TTATATAA

GIBBS IC

10.66 bits

GIBBS SCORE

158.37**MEME Sequence Logo****Gibbs Sequence Logo****PWM Heatmap****Gibbs Convergence****Site Position Distribution (MEME)**

Synthetic E-box

Planted CACGTG benchmark

MEME CONSENSUS

ACGTGT

MEME IC

6.90 bits

E-VALUE

1.42e-76

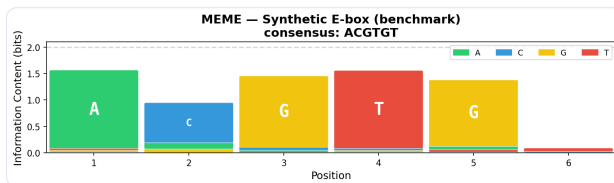
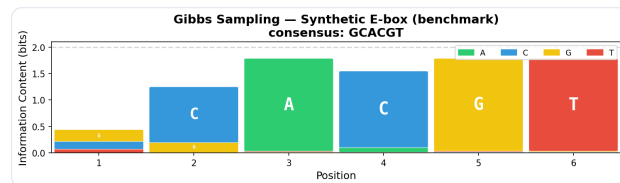
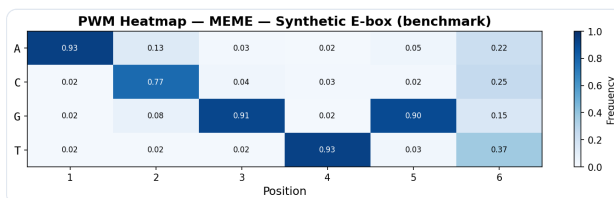
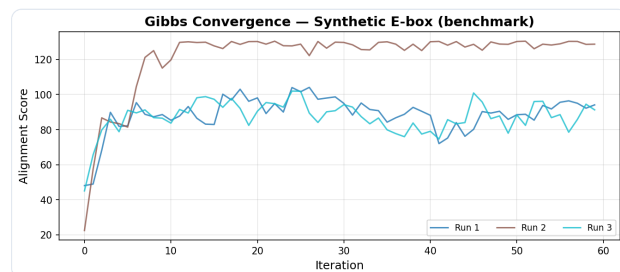
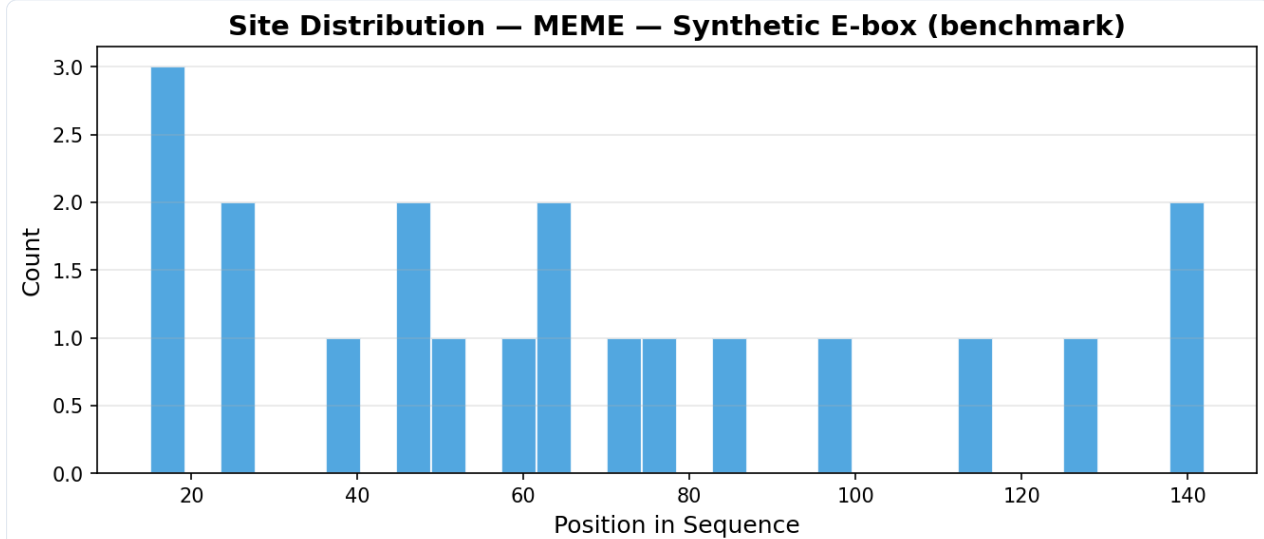
GIBBS CONSENSUS

GCACGT

GIBBS IC

8.73 bits

GIBBS SCORE

130.42**MEME Sequence Logo****Gibbs Sequence Logo****PWM Heatmap****Gibbs Convergence****Site Position Distribution (MEME)**

Real Data Provenance (JASPAR CORE)

All PFMs fetched live from <https://jaspar.elixir.no/api/v1/> on 20 April 2026. Each entry is backed by the listed PubMed publication.

MATRIX ID	TF NAME	ORGANISM	TAX GROUP	EXPERIMENT	PUBMED	WIDTH	CONSENSUS
MA0001.1	AGL3	<i>Arabidopsis thaliana</i>	plants	SELEX	7632923	10	CCATAAATAG
MA0570.1	ABF1_plant	<i>Arabidopsis thaliana</i>	plants	SELEX	10636868	19	GGACACGTGGCAGCAGGG
MA0941.1	ABF2_plant	<i>Arabidopsis thaliana</i>	plants	PBM	11005831	13	CATGACACGTGTT
MA0930.1	ABF3_v1	<i>Arabidopsis thaliana</i>	plants	PBM	11005831	8	ACACGTGT
MA0930.2	ABF3_v2	<i>Arabidopsis thaliana</i>	plants	ChIP-seq	11005831	10	CACGTGTCAT
MA0930.3	ABF3_v3	<i>Arabidopsis thaliana</i>	plants	ChIP-seq	11005831	7	ACGTGTC
MA1659.1	ABF4_v1	<i>Arabidopsis thaliana</i>	plants	DAP-seq	25220462	12	TGCCACGTCACC
MA1659.2	ABF4_v2	<i>Arabidopsis thaliana</i>	plants	DAP-seq	25220462	7	CCACGTC
MA0564.1	ABI3_v1	<i>Arabidopsis thaliana</i>	plants	PBM	30183137	9	CTGCATGCA
MA0564.2	ABI3_v2	<i>Arabidopsis thaliana</i>	plants	PBM	30183137	6	GCATGC
MA0931.1	ABI5_v1	<i>Arabidopsis thaliana</i>	plants	PBM	12376636	10	TGACACGTGG
MA0931.2	ABI5_v2	<i>Arabidopsis thaliana</i>	plants	PBM	12376636	8	GACACGTG
MA1244.1	ABR1	<i>Arabidopsis thaliana</i>	plants	DAP-seq	9756931	19	GGCGGAAATGGCGGCGGG
MA0265.1	ABF1_yeast	<i>Saccharomyces cerevisiae</i>	fungi	PBM/DIP-chip	19111667	16	TCGTATAAAGTGATAA
MA0266.1	ABF2_yeast_v1	<i>Saccharomyces cerevisiae</i>	fungi	PBM	19111667	7	CTCTAGA
MA0266.2	ABF2_yeast_v2	<i>Saccharomyces cerevisiae</i>	fungi	PBM	19111667	6	TCTAGA
MA0139.2	CTCF	<i>Homo sapiens</i>	vertebrates	ChIP-seq	21106759	15	GCCACCAGGGGGCGC
MA0024.3	E2F1	<i>Homo sapiens</i>	vertebrates	HT-SELEX	9372931	12	TTTGCGCCAAA
MA0024.2	E2F1_chipseq	<i>Homo sapiens</i>	vertebrates	ChIP-seq	17908821	11	CGGGCGGGAGG

MATRIX ID	TF NAME	ORGANISM	TAX GROUP	EXPERIMENT	PUBMED	WIDTH	CONSENSUS
MA0899.1	HOXA10	<i>Homo sapiens</i>	vertebrates	HT-SELEX	18585359	11	GGTAATAAAAA
MA0020.2	Dof2_maize	<i>Zea mays</i>	plants	SELEX	10074718	5	AAAGC
MA0021.1	Dof3_maize	<i>Zea mays</i>	plants	SELEX	10074718	5	AAAGC